

Abstract

Background:

Artificial intelligence (AI) appears capable of detecting diabetic retinopathy (DR) with a high degree of accuracy in adults; however, there are few studies in children and young adults.

Methods:

Children and young adults (3-26 years) with type 1 diabetes mellitus (T1DM) or type 2 diabetes mellitus (T2DM) were screened at the Dhaka BIRDEM-2 hospital, Bangladesh. All gradable fundus images were uploaded to Cybersight AI for interpretation. Two main outcomes were considered at a patient level: 1) Any DR, defined as mild non-proliferative diabetic retinopathy (NPDR or more severe; and 2) Referable DR, defined as moderate NPDR or more severe. Diagnostic test performance comparing Orbis International's Cybersight AI with the reference standard, a fully qualified optometrist certified in DR grading, was assessed using the Matthews correlation coefficient (MCC), area under the receiver operating characteristic curve (AUC-ROC), area under the precision-recall curve (AUC-PR), sensitivity, specificity, positive and negative predictive values.

Results:

Among 1274 participants (53.1% female, mean age 16.7 years), 19.4% (n = 247) had any DR according to AI. For referable DR, 2.35% (n = 30) were detected by AI. The sensitivity and specificity of AI for any DR were 75.5% (CI 69.7-81.3%) and 91.8% (CI 90.2-93.5%) respectively, and for referable DR, these values were 84.2% (CI 67.8-100%) and 98.9% (CI 98.3%-99.5%). The MCC, AUC-ROC and the AUC-PR for referable DR were 63.4, 91.2 and 76.2% respectively. AI was most successful in accurately classifying younger children with shorter duration of diabetes.

Conclusions:

Cybersight AI accurately detected any DR and referable DR among children and young adults, despite its algorithms having been trained on adults. The observed high specificity is particularly important to avoid over-referral in low-resource settings. AI may be an effective tool to reduce demands on scarce physician resources for the care of children with diabetes in low-resource settings.

Introduction

Diabetes mellitus (DM) is one of the largest global health threats of the 21st century.¹ The prevalence of DM has been rising more rapidly in low- and middle-income countries (LMICs) compared to high-income countries.¹ The International Diabetes Federation (IDF) reports that 3 out of 4 people living with DM dwell in LMICs.¹ A particularly concerning and less-studied facet of the global DM epidemic is the rapidly growing prevalence among children and young adults. Globally, an estimated 1.1 million children and adolescents (<20 years) are living with type 1 DM (T1DM).¹ In countries with poor health systems and limited access to insulin, complications are common in young people with T1DM.¹ Additionally, lifestyle changes across both high and low-income countries have led to a rise in obesity, and consequently type 2 DM (T2DM) among children and young adults.² The World Health Organization (WHO) reports that until recently, T2DM was only seen in adults, but is now occurring with increasing frequency in children.³ According to IDF estimates in 2020, around 44 million children and adolescents (0-19 years) had T2DM globally.¹ The prevalence of T2DM among children and adolescents varies across countries and regions; however, it is more prevalent in regions with a high burden of obesity and lifestyle related risk factors.^{1,4} There is a lack of information on T1DM/T2DM, particularly in LMICs, where incidence remains largely unknown, further research and diabetes registries are required to address these gaps.

Bangladesh is a LMIC in South Asia with one of the fastest-growing economies globally.⁵ Unfortunately, the prevalence of DM is showing the largest increases in countries with high economic growth.⁶ Zabeen et al⁷ reported that the incidence of T2DM per 100 000 young people in Bangladesh aged <20 years nearly tripled from 0.2 to 0.57 between 2011 and 2018, a 12% increase per year ($P < .001$). This is particularly concerning given that early-onset T2DM exhibits accelerated development of systemic complications.⁸

Diabetic retinopathy (DR) is one of the most common microvascular complications of DM, and can lead to vision impairment and blindness if not detected and treated early.⁹ Regular DR screening and treatment reduces severe vision loss up to 90%.¹⁰ While the prevalence of DR among youths with well-controlled T1DM is low, adolescents with T2DM have a higher risk of DR progression compared to adults, especially when glycaemic control is poor.¹